ECHOES: EQUAL-WEIGHT COMPLETED HYPOTHETICAL OBSERVATION ENSEMBLES FOR BOSS DR12 CMASS-SOUTH: POSTERIOR COMPLETED-CATALOG SAMPLES FOR GALAXY CLUSTERING BEYOND TWO POINTS

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ABSTRACT

Most precision galaxy-clustering analyses communicate between survey teams and theory teams through a small set of summary statistics, most often two-point correlation functions or power spectra. This convention is effective because the observational effects that enter these statistics have been studied for decades, but it also makes each new statistic responsible for reinterpreting the survey window, fiber assignment, redshift failures, imaging systematics, and random catalogs. We present ECHOES, Equal-weight Completed Hypothetical Observation Ensembles, for BOSS DR12 CMASS-South: an ensemble of equal-weight, cosmology-free completed catalogs that keep every observed galaxy at its measured (α, δ, z) and restore the spectroscopically missing galaxies at their real SDSS imaging positions. The unknown redshifts are sampled from a data-driven posterior combining a color-space k-nearest-neighbor photo-z, an empirical close-pair prior, and a local-density line-of-sight prior. Smooth imaging-systematic weights are represented by local analog galaxies rather than by per-object completeness weights. The released posterior is compact: one 2.04 MB file stores the fixed observed catalog and the inverse CDF for each restored target, and a numpy-only sampler generates reproducible complete catalogs from integer seeds. For seed 0 the sample contains 119,923 equal-weight galaxies, of which 109,636 are observed, 5,272 are restored fiber-collision targets, 1,505 are restored redshift failures, and 3,510 are imaging-systematic analogs. We validate the product against truth-known completions and MultiDark-Patchy mocks, finding that projected clustering is recovered to the 1–2% level over $0.5 < r_p < 40\,h^{-1}$Mpc, while the posterior completion uncertainty in $w_p(r_p)$ is only 0.17–0.44% per bin, below the sample-variance scale of the BOSS South Galactic Cap. We also describe a complementary field-level route: a kernel (FKP) estimate of the galaxy density field, built with a covariance tabulated from the measured correlation function and sampled as a coverage-scaled ensemble on a sparse neighbor graph, evaluated along each missing galaxy's line of sight. The resulting catalogs are intended to let analysts run two-point, higher-order, marked, nearest-neighbor, topological, field-level, or simulation-based statistics directly on completed catalogs while propagating the observational-completion uncertainty by resampling the catalog posterior.

Subject headings: Large-scale structure of the universe (902); Galaxy surveys (598); Astrostatistics (1882); Cosmological parameters (339); Observational cosmology (1146)

1. INTRODUCTION

The central compression in observational large-scale structure is not a simulation or an estimator, but an agreement about what object is being communicated. A survey team measures a summary statistic from a complicated instrumental and astrophysical selection function; a theory team predicts the same statistic for a cosmological and galaxy-formation model. The summary statistic is the interface. For modern spectroscopic galaxy surveys that interface has usually been a two-point statistic: a correlation function, a projected correlation function, a redshift-space multipole, or a Fourier-space power spectrum (M. Davis & P. J. E. Peebles 1983; S. D. Landy & A. S. Szalay 1993; A. J. S. Hamilton 1993; H. A. Feldman et al. 1994). This choice is not merely historical. The observational corrections for two-point clustering have been revisited repeatedly, calibrated in mocks, and encoded in survey catalogs, random catalogs, and weights.

At the same time, the cosmological information in a galaxy survey is not exhausted by two-point clustering. The literature contains a long sequence of proposed and demonstrated summaries: higher-order moments and hierarchical statistics (I. Szapudi & A. S. Szalay 1993; J. N. Fry & E. Gaztanaga 1993), void-probability functions (J. Einasto et al. 1991), counts-in-cells and one-point density PDFs (C. M. Baugh et al. 1995; I. Szapudi et al. 1996; C. Uhlemann et al. 2020; O. Friedrich et al. 2020), marked correlation functions and marked power spectra (G. Harker et al. 2006; S. Satpathy et al. 2019; E. Massara et al. 2021, 2023), nearest-neighbor distributions (A. Banerjee & T. Abel 2021; S. Yuan et al. 2024), wavelet and scattering-transform statistics (S. Cheng et al. 2020; G. Valogiannis & C. Dvorkin 2022; B. Régaldo-Saint Blancard et al. 2024), persistent-homology summaries of the cosmic web (G. Wilding et al. 2021; M. Biagetti et al. 2022; J. H. T. Yip et al. 2024), and field-level or simulation-based inference methods (J. Jasche & B. D. Wandelt 2013; N.-M. Nguyen et al. 2024; P. Lemos et al. 2024; C. Hahn et al. 2023a,b, 2024a,b; E. Massara et al. 2025). Large simulation suites such as Quijote (F. Villaescusa-Navarro et al. 2020) have been essential for calibrating and stress-testing many of these

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summaries. The Beyond-2pt Collaboration's parameter-masked mock-data challenge (ThBeyond-2pt Collaboration et al. 2025) makes the same point in a deliberately broad form: once the Gaussian, two-point description is relaxed, many summaries can extract complementary information, and their relative utility is an empirical question.

The barrier between those ideas and real spectroscopic data is often not the statistic itself. It is the observational model. A statistic that is easy to define on a periodic simulation box can become difficult to interpret in a survey with angular holes, variable targeting, fiber collisions, redshift failures, stellar density, seeing, extinction, and redshift-dependent selection. For two-point statistics, decades of work have made the corresponding correction pathways usable. For many newer statistics, the same corrections are either rederived from scratch, approximated, or avoided by restricting the analysis to mocks and forecasts. This paper asks whether the interface can be moved one step earlier: instead of releasing only weighted galaxies and randoms and asking every statistic to understand the weights, release a posterior over completed galaxy catalogs that have already absorbed the main observational incompleteness into equal-weight point sets.

This perspective has a long historical lineage. The first large maps of extragalactic structure already had to confront the fact that the sky is not a uniform observing surface. E. Hubble (1934) discussed the distribution of extragalactic nebulae in the presence of Galactic obscuration, a problem that later became a quantitative dust-correction program culminating in all-sky reddening maps such as D. J. Schlegel et al. (1998). Redshift surveys then turned angular inhomogeneity into a three-dimensional selection problem (M. Davis & P. J. E. Peebles 1983; V. de Lapparent et al. 1986). The 2dF Galaxy Redshift Survey and SDSS established the modern industrial mode of galaxy redshift surveys (D. G. York et al. 2000; M. Colless et al. 2001; S. Cole et al. 2005), and the SDSS luminous red galaxy sample made the two-point correlation function a precision cosmological observable with the detection of the baryon acoustic feature (D. J. Eisenstein et al. 2005).

BOSS inherited this history and made it operational. Its spectrographs, fiber system, target selection, and large-scale-structure catalogs are documented in K. S. Dawson et al. (2013); S. A. Smee et al. (2013); B. Reid et al. (2016). The final DR12 cosmology analyses used a mature system of angular masks, radial random catalogs, redshift-failure weights, close-pair weights, imaging-systematic weights, and FKP weights (S. Alam et al. 2017; A. J. Ross et al. 2017). The resulting measurements were robust because the effect of these ingredients on the two-point statistics had been studied. But the same survey artifacts have no reason to couple only to two-point functions. Fiber collisions remove close angular pairs; redshift failures correlate with observing conditions and target properties; imaging systematics modulate the target density across degree scales; and the random catalog carries a particular representation of the angular and radial selection function.

The same issues will not become simpler in later surveys. DESI's robotic fiber-positioner system and its fiber-assignment algorithm create a richer assignment problem than the BOSS plug-plate geometry (J. H. Silber et al. 2023; L. Pinol et al. 2017; T.-W. Lan et al. 2023; J. Lasker et al. 2025). Euclid, the Nancy Grace Roman Space Telescope, the Chinese Space Station Telescope, and the Vera C. Rubin Observatory's LSST will produce enormous galaxy samples with their own mixtures of imaging, spectroscopy, selection, and calibration systematics (Euclid Collaboration et al. 2025; Y. Wang et al. 2022; H. Miao et al. 2023; CSST Collaboration et al. 2026; H. Zhan & J. A. Tyson 2018). If those surveys wish to serve a community that will analyze the data with statistics not yet standardized, the most useful data product may not be a single corrected summary statistic. It may be a completed probabilistic catalog.

This work builds such a product for BOSS DR12 CMASS-South. We call the approach ECHOES: Equal-weight Completed Hypothetical Observation Ensembles, echoes of the survey one would have taken if the principal spectroscopic incompleteness and smooth imaging systematics were negligible. It is not a new cosmological measurement. It is a data-modeling and validation paper. We construct an ensemble of completed catalogs conditioned on the published BOSS data and SDSS imaging: observed spectroscopic galaxies remain fixed, missing targets are restored at real imaging positions, and redshift uncertainty is sampled. The ensemble is cosmology-free in the sense that it contains only RA, Dec, and redshift; an analyst chooses a fiducial cosmology only when measuring three-dimensional clustering. The ECHOES product is designed so that the summary statistic, rather than the observational correction, can once again be the object under study.

2. ECHOES COMPLETED-SURVEY METHODOLOGY

2.1. *Data model and scope*

The target ECHOES product is an ensemble

$$\mathcal{C}^{(s)} = \{\alpha_i^{(s)}, \delta_i^{(s)}, z_i^{(s)}, p_i^{(s)}\}_{i=1}^{N_s}, \tag{1}$$

where s is a seed and p_i is a provenance code. The coordinates are observed RA, Dec, and redshift. No comoving coordinate, Alcock–Paczynski scaling (C. Alcock & B. Paczynski 1979), redshift-space model (N. Kaiser 1987), or fiducial cosmology is stored in the catalog. The completion posterior is conditional on the observed BOSS catalog, the SDSS imaging target catalog, and the BOSS angular mask and weights. It does not represent cosmic variance; it represents the uncertainty in replacing spectroscopic incompleteness and smooth imaging-systematic weights by equal-weight galaxies.

The ECHOES product corrects three effects in the default release: fiber collisions, redshift failures, and the smooth imaging-systematic modulation represented by `WEIGHT_SYSTOT`. An optional inpainted version also fills interior mask holes, but the default scientific product leaves the survey mask explicit. The FKP weight (H. A. Feldman et al. 1994) is not a completeness correction and is not converted into galaxies. Analysts may still apply FKP or other inverse-variance weights when estimating a statistic; the released point catalog itself is equal weight.

2.2. *BOSS CMASS-South sample*

We use the BOSS DR12 CMASS South Galactic Cap subset with the same clean cuts used in several modern analyses of these data:

$$\mathrm{RA} < 28^\circ \text{ or } \mathrm{RA} > 335^\circ, \qquad \mathrm{Dec} > -6^\circ, \qquad 0.45 < z < 0.60. \tag{2}$$

The input contains 109,636 observed spectroscopic galaxies. The BOSS completeness weights imply a mean completeness correction of 1.074, or about 7.4% missing objects before the smooth imaging-systematic analog step. In the real target recovery used here, 5,272 fiber-collision targets and 1,505 redshift-failure targets are restored, for 6,777 spectroscopically missing galaxies in the compact posterior package.

The BOSS loader reads the standard large-scale-structure columns `RA`, `DEC`, `Z`, `WEIGHT_SYSTOT`, `WEIGHT_NOZ`, `WEIGHT_CP`, and `WEIGHT_FKP`. It also reads `MODELFLUX`, `EXTINCTION`, `IMATCH`, and `ICOLLIDED` when photometry is requested. Extinction-corrected Pogson magnitudes are computed as

$$m_b = 22.5 - 2.5 \log_{10} f_b - A_b, \tag{3}$$

with adjacent colors $(u-g, g-r, r-i, i-z)$. The photometric-redshift model uses $(g-r, r-i, i-z, i)$, dropping the u band because it is noisy for the red CMASS population.

2.3. *Recovering the missing target list*

The completion uses real imaging positions for missing galaxies. We start from the DR12 CMASS color-selected target catalog, cross-matched to the good redshift BOSS LSS catalog. A target with a spectrum but nonzero `ZWARNING` is classified as a redshift failure. A target with no usable spectrum is eligible to be a fiber-collision target. Because the no-spectrum photometric pool also contains objects that were not part of the final BOSS LSS sample, it is tied back to the BOSS bookkeeping. For each observed survivor with `WEIGHT_CP`> 1, the algorithm greedily claims round(`WEIGHT_CP`-1) nearest unclaimed no-spectrum targets within the BOSS fiber-collision scale, $62''$. Redshift failures are tied to survivors with `WEIGHT_NOZ`> 1 using the analogous round(`WEIGHT_NOZ` $- 1$) count and a wider $15'$ neighborhood. This preserves the integer count implied by the official weights while placing the restored galaxies at their actual imaging positions and colors.

2.4. *Color-space photo-z posterior*

For each missing target we require a redshift posterior, not only a point estimate. We fit a nonparametric k-nearest-neighbor model to the observed CMASS galaxies in the whitened four-dimensional feature space $(g-r, r-i, i-z, i)$. The default model uses $k = 100$ and a `cKDTree`. For a query feature vector $\mathbf{x}$, the posterior support is the set of neighbor redshifts $\{z_j\}_{j=1}^{k}$ with inverse-distance weights

$$w_j(\mathbf{x}) = \frac{(d_j^2 + \epsilon)^{-1}}{\sum_{\ell=1}^{k} (d_\ell^2 + \epsilon)^{-1}}. \tag{4}$$

The held-out diagnostic in the report gives $\sigma_{\mathrm{NMAD}} = 0.019$, a bias of -0.0006, a catastrophic outlier rate of 1.8%, and a probability-integral-transform mean of 0.503. These numbers are adequate for the role the photo-z plays here: it supplies a color likelihood that is then multiplied by a local-density prior along the same line of sight.

2.5. *Local-density redshift posterior*

The default redshift engine is a fast local-density approximation to a conditional field posterior. For missing target i at angular position $\hat{\mathbf{n}}_i$ and colors $\mathbf{x}_i$, redshifts are sampled from

$$p_i(z) \propto \widehat{\lambda}_{\mathrm{loc}}(\hat{\mathbf{n}}_i, z)\, \widehat{p}_{\mathrm{col}}(z \mid \mathbf{x}_i)\, q_i(z), \tag{5}$$

where $\widehat{p}_{\mathrm{col}}$ is the color-space photo-z density, q_i is an empirical close-pair prior for collisions and unity for redshift failures, and $\widehat{\lambda}_{\mathrm{loc}}$ is the observed local line-of-sight density. The local density is estimated from the $K = 150$ nearest observed spectroscopic galaxies in angular unit-vector space:

$$\widehat{\lambda}_{\mathrm{loc}}(\hat{\mathbf{n}}_i, z) = \sum_{j \in \mathcal{N}_K(i)} \exp\left[-\frac{(z - z_j)^2}{2\sigma_f^2}\right], \qquad \sigma_f = 0.004. \tag{6}$$

The photo-z factor is evaluated on the same 256-point redshift grid with a Gaussian smoothing scale $\sigma_p = 0.02$ applied to the neighbor redshifts and their kNN weights. For collided targets, $q_i(z)$ is the symmetrized histogram of redshift differences among observed close pairs within $62''$, measured directly from the data over $|\Delta z| < 0.06$. The sampled redshift is then jittered by a Gaussian of width $\sigma_f/2$ and clipped to the CMASS redshift interval.

Equation (5) is deliberately observational. It does not assign a missing galaxy to a simulated halo or require a fiducial cosmology. The object is placed at its real angular position and on a redshift structure seen in the neighboring spectroscopy, with the color likelihood preventing the local-density prior from assigning obviously incompatible redshifts.

2.6. *Imaging-systematic analogs and provenance*

The BOSS imaging-systematic correction `WEIGHT_SYSTOT` is a smooth density correction, not a statement that a particular nearby target failed to receive a fiber. We therefore do not convert it into exact duplicate galaxies. The completion is *mean-preserving*: the expected number of output objects per base object equals w_{systot}. For every observed or spectroscopically restored base object, the sampler adds

$$n_{\mathrm{extra}} = \lfloor \max(w_{\mathrm{systot}} - 1, 0) + U \rfloor, \qquad U \sim \mathcal{U}(0,1), \tag{7}$$

local analogs at the same source position with a $1''$ Gaussian angular jitter and the same redshift, and *independently keeps the base object itself with probability* $\min(w_{\mathrm{systot}}, 1)$. The analog step restores the density deficit where $w_{\mathrm{systot}} > 1$; the keep step thins the regions with $w_{\mathrm{systot}} < 1$, which for CMASS-South are the majority (median $w_{\mathrm{systot}} = 0.98$, 64% below unity). Restoring the deficit without thinning the excess would reproduce a $\max(w_{\mathrm{systot}}, 1)$ weighting and leave a residual degree-scale imaging-systematic gradient — in CMASS-South the $w_{\mathrm{systot}} < 1$ regions would be over-dense by up to $\sim 8\%$ (Figure 3); the thinning removes it to $\lesssim 0.2\%$. The jitter preserves resolved angular and three-dimensional

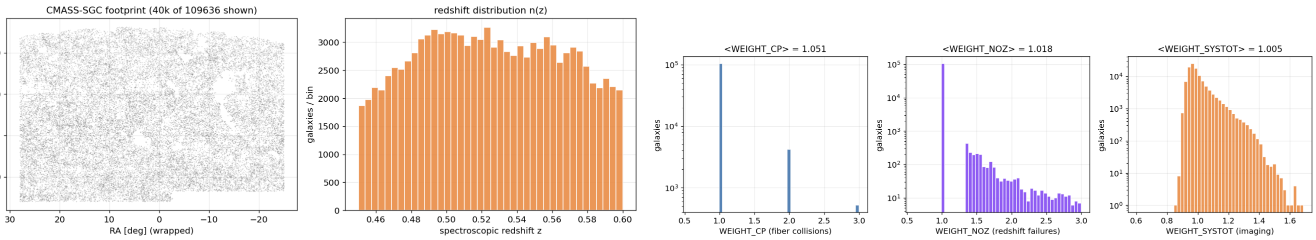

FIG. 1.— Inputs to the completion. *Left:* the BOSS DR12 CMASS-South footprint and observed-galaxy distribution on the sky. *Right:* the published completeness weights — fiber-collision $w_{\rm cp}$, redshift-failure $w_{\rm noz}$, and imaging-systematic $w_{\rm systot}$ — whose product defines the per-region incompleteness the completion restores. The completion is conditioned on this geometry, on these weights, and on the SDSS imaging positions of the spectroscopically missing targets; it converts the weights into explicit equal-weight galaxies rather than carrying them as per-object multipliers.

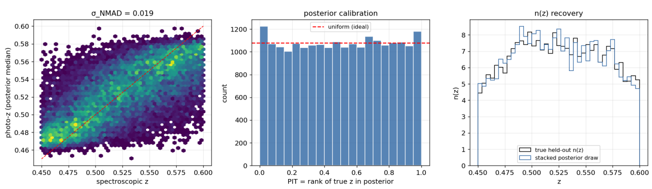

FIG. 2.— The color-space k-nearest-neighbor photo-z that supplies the color likelihood in the redshift posterior, validated on a held-out sample of good-redshift CMASS galaxies. *Left:* posterior-median photo-z versus spectroscopic redshift ($\sigma_{\rm NMAD} \simeq 0.019$). *Middle:* the probability-integral-transform (PIT) histogram — the rank of each true redshift within its own posterior — which is statistically uniform, so the posterior is calibrated and can be sampled faithfully. We test uniformity with a Kolmogorov–Smirnov and a binned χ^2 statistic against the flat distribution rather than the PIT mean alone, since a U-shaped (over-confident) PIT can also have mean 0.5. *Right:* a single posterior draw per object reproduces the held-out $n(z)$. The photo-z enters the completion only as the likelihood factor $\widehat{p}_{\rm col}(z \mid \mathbf{x})$, never as a standalone redshift.

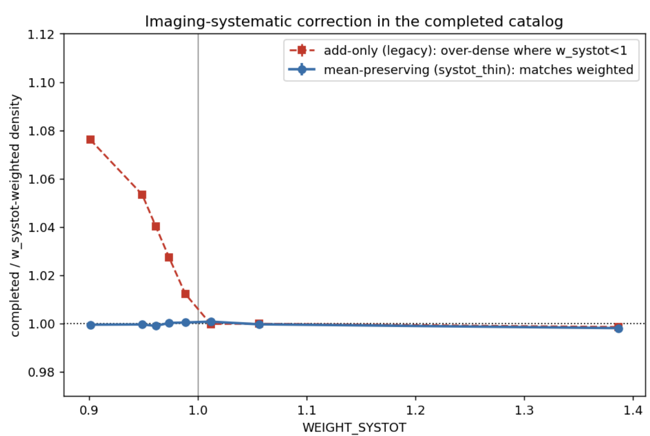

FIG. 3.— Mean-preserving imaging-systematic completion. The completed number density per $w_{\rm systot}$ bin, divided by the $w_{\rm systot}$-weighted target density (isolating the systematic step on real CMASS-South). The mean-preserving completion (blue) reproduces the weighted density to $\lesssim 0.2\%$ across the full range, whereas restoring the deficit without thinning the $w_{\rm systot} < 1$ regions (red) leaves them over-dense by up to $\sim 8\%$, a residual degree-scale gradient correlated with the imaging systematic.

clustering while avoiding a zero-separation spike that would corrupt nearest-neighbor or counts-in-cells statistics. The keep/drop is stochastic per realization, so the $w_{\rm systot}$-weighted density is reproduced in the ensemble mean and the small per-draw thinning shot noise enters the calibrated completion spread.

Each output object carries a provenance code: 0 for observed spectroscopy, 1 for restored fiber-collision target, 2 for restored redshift failure, 3 for imaging-systematic analog, 4 for rare host-redshift fallback, and 5 for optional mask inpainting. The provenance vector is central to reproducibility because analysts can rerun a statistic on individual components or test sensitivity to the imaging analog prescription.

2.7. *Compact posterior and sampling*

The default posterior has a simple product structure once the observed data are fixed. The observed galaxies never change across realizations. The missing galaxy positions never change. The redshift posterior for each missing galaxy, Equation (5), is also fixed after it is computed. A realization is only a draw from each per-object inverse CDF plus the stochastic imaging analogs.

The release therefore stores this posterior as a single compressed file of 2.04 MB: the fixed observed catalog, the base positions and provenance for the observed and restored targets, `WEIGHT_SYSTOT`, and a tabulated inverse-CDF redshift posterior for each of the 6,777 missing targets (the file schema and quantization are given in Appendix C). A matching uniform-footprint random catalog contains 438,544 points; because CMASS-South is approximately 99% complete in the angular mask, these randoms are uniform over the footprint to the percent level.

Sampling is deterministic under a seed:

1. draw one uniform variate for each missing target and invert its stored redshift CDF;
2. concatenate the fixed observed galaxies and the restored targets;
3. draw local analog counts from `WEIGHT_SYSTOT`;
4. return arrays `ra`, `dec`, `z`, `prov`, and `N`.

For seed 0, the sampled catalog contains 119,923 objects: 109,636 observed, 5,272 collision restorations, 1,505 redshift-failure restorations, and 3,510 smooth-systematic analogs. Without the imaging analogs, the fixed base catalog contains 116,413 objects.

2.8. *Clustering measurements and acceptance metrics*

For angular clustering we measure Landy–Szalay estimates (S. D. Landy & A. S. Szalay 1993) with the fixed random catalog. For three-dimensional statistics we use `Corrfunc` (M. Sinha & L. H. Garrison 2020). The catalog itself is cosmology-free; `Corrfunc` measurements convert redshift to comoving distance at measurement time, using the analyst's fiducial cosmology,

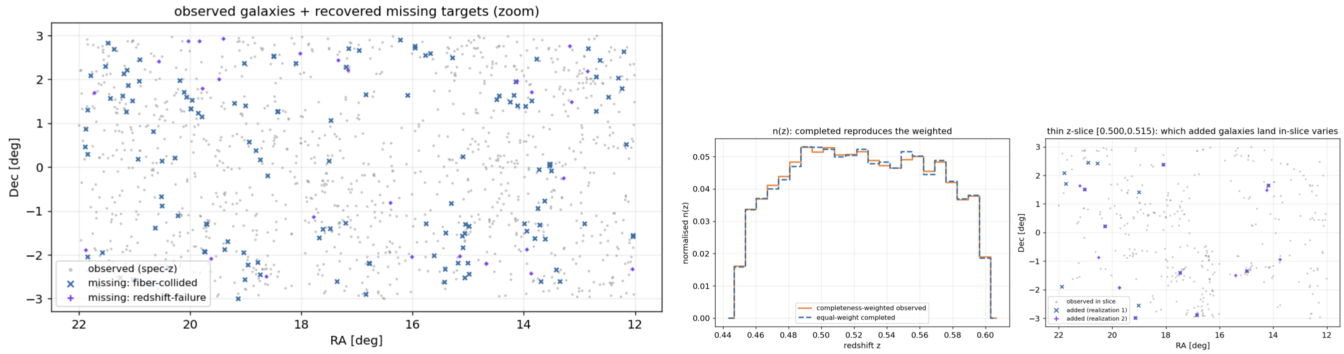

FIG. 4.— Restored missing targets and the resulting redshift distribution. *Left:* a zoomed sky region showing observed galaxies together with the recovered fiber-collision and redshift-failure targets, each placed at its real SDSS imaging position; only the redshift of each restored galaxy is uncertain. *Right:* the equal-weight completed $n(z)$ reproduces the completeness-weighted observed $n(z)$, and the thin-slice panel illustrates that *which* restored galaxies fall in a given redshift slice varies between realizations — this realization-to-realization variation is the completion posterior. Realizations differ only in the sampled redshifts of the missing galaxies and in the stochastic imaging-systematic analogs.

and then call the OpenMP mock pair counters with `is_comoving_dist=True`. The main validation statistics are $w(\theta)$, $w_p(r_p)$, $\xi(r_p, \pi)$, and redshift-space multipoles $\xi_0(s)$ and $\xi_2(s)$. We also evaluate higher-order or non-two-point summaries: nearest-neighbor CDFs and counts-in-cells. These are deliberately included because the purpose of the catalog posterior is to support statistics beyond those for which BOSS weights were optimized.

The acceptance criteria are empirical. In truth-known tests, the completed ensemble should recover the truth catalog to the percent level over the scales where the input mocks themselves are faithful. On the real data, the equal-weight completed catalogs should reproduce the official $w_c = \texttt{WEIGHT\_SYSTOT}\,(\texttt{WEIGHT\_CP} + \texttt{WEIGHT\_NOZ} - 1)$ weighted BOSS clustering within the known scale of the correction. The completion posterior spread should be smaller than the sample-variance covariance for BOSS CMASS-South, because it represents only uncertainty in observational completion conditioned on the observed catalog.

3. THE GRAPHGP COMPLEMENT

The local-density engine in Section 2.5 is fast and compressible, but it treats the missing-galaxy redshift posteriors as conditionally independent once the observed catalog is fixed. The graphGP route is a complementary, more explicit field-level model. It assumes that the complete galaxy overdensity field can be approximated by a Gaussian process with covariance tied to the observed two-point function, and that the observed galaxies are noisy samples of that field.

Let $\delta(\mathbf{x})$ be the galaxy overdensity in a fiducial comoving coordinate system. The clustering scale is set by a kernel

$$K_\xi(r) \text{ tabulated from } \mathcal{GP}\left[0, K_\xi(|\mathbf{x} - \mathbf{x}'|)\right], \qquad (8)$$

a covariance derived from a Landy–Szalay measurement of the galaxy correlation function, so the smoothing follows the measured two-point clustering rather than an imposed bandwidth. In the regime relevant here the signal-to-noise per galaxy is small ($\bar{n}_i^{-1} \gg \xi$), and we use the kernel-weighted (FKP) density-ratio estimate of the field — the posterior-mean limit of the Gaussian model when the Poisson noise dominates the prior:

$$1 + \delta_{\rm FKP}(\mathbf{x}) = \frac{\sum_i w_i\, K_\xi(|\mathbf{x} - \mathbf{x}_i^D|)}{\alpha_w \sum_j K_\xi(|\mathbf{x} - \mathbf{x}_j^R|)}, \qquad \alpha_w = \frac{\sum_i w_i}{N_R}, \qquad (9)$$

normalized to unit mean over the survey, with completeness weights w_i and the random catalog $\mathbf{x}^R$ supplying the angular$\times$radial selection in the denominator. An ensemble of field realizations is then generated by adding a coverage-scaled stochastic term,

$$\delta^{(s)}(\mathbf{x}) = \delta_{\rm FKP}(\mathbf{x}) + s_0\,[1 - c(\mathbf{x})]\,\eta^{(s)}(\mathbf{x}), \qquad (10)$$

where $c(\mathbf{x}) \in [0, 1]$ is the local data proximity, s_0 calibrates the amplitude to the measured field, and $\eta^{(s)}$ is a prior draw — generated on a sparse Vecchia-style neighbor graph (M. Katzfuss et al. 2020; F. Schäfer et al. 2021) so it carries the ξ-correlation at the data positions. The spread is therefore full-amplitude where the survey is data-poor and suppressed near data, the qualitative behavior of a field posterior, without forming the $[K_{DD} + N]^{-1}$ conditional solve. The primary stored object is a light-cone-native array $1 + \delta^{(s)}$ on redshift shells and HealPIX pixels (the stored noise floor is uncorrelated between voxels). For a missing galaxy, this route replaces the local KDE in Equation (5) by the corresponding field draw:

$$p_i(z \mid s) \propto [1 + \delta^{(s)}(\hat{\mathbf{n}}_i, z)]\, \bar{n}(z)\, \widehat{p}_{\rm col}(z \mid \mathbf{x}_i)\, q_i(z). \qquad (11)$$

All missing galaxies in realization s share the same density-field estimate, so their redshift assignments are correlated through the smooth shared field; the stored light-cone noise term is uncorrelated between voxels. We emphasize that this engine is a fast, robust field-level *estimate* (an FKP kernel field with a coverage-scaled noise ensemble), not an exact Gaussian-process conditional posterior; it is the optional complement to the default KNN-field route, which backs the released product.

This model has a fiducial comoving coordinate system internally because a three-dimensional covariance needs a distance. That coordinate system is a modeling gauge for the field interpolation, not a cosmology baked into the released KNN product. The report tests this explicitly by rebuilding the graphGP route with different fiducial distance-redshift relations. Planck-like and Einstein–de

Sitter choices give per-object redshift RMS differences of about 0.001–0.0012, correlations of 0.9995–0.9996, and changes in $w_p(r_p)$ below 0.1%. Within the accuracy demanded by the present BOSS completion, graphGP is therefore best viewed as a correlated posterior engine whose output is stable to reasonable internal coordinate choices.

4. APPLICATION TO BOSS DR12 CMASS-SOUTH

4.1. *Why BOSS is the right test case*

BOSS DR12 is the ideal laboratory because the official two-point analyses are mature. The survey, target selection, and LSS catalogs are documented (K. S. Dawson et al. 2013; B. Reid et al. 2016), the spectroscopic system is understood (S. A. Smee et al. 2013), the DR12 cosmology and systematic-correction papers provide community reference measurements (S. Alam et al. 2017; A. J. Ross et al. 2017), and independent mock suites such as MultiDark-Patchy exist for validation (F.-S. Kitaura et al. 2016). A completed-catalog product can therefore be judged against a well-established reference: it should reproduce the weighted BOSS two-point measurements while exposing a point catalog suitable for statistics that were not part of the official BOSS pipeline.

The official BOSS correction strategy keeps the observed spectroscopic catalog and supplies per-object weights. Close-pair and redshift-failure weights move the missing object's statistical weight to an observed neighbor, while `WEIGHT_SYSTOT` corrects smooth density modulations associated with imaging and observing conditions. Random catalogs encode the angular mask and radial selection. This system is extremely effective for the published two-point program, but for a general statistic the analyst must decide how to apply weights, whether the statistic is linear or nonlinear in those weights, and how random catalogs should enter. Fiber-collision corrections illustrate the point. Nearest-neighbor and angular upweighting are calibrated for two-point functions (H. Guo et al. 2012), but the more powerful pairwise-inverse-probability (PIP), or bitwise, weighting of D. Bianchi & W. J. Percival (2017), validated on BOSS and eBOSS by F. G. Mohammad et al. (2020), is in fact *statistic-agnostic*: by reweighting pairs with the inverse probability that they are jointly observed across realizations of the targeting algorithm, it yields unbiased clustering for a general pair counter, not only the two-point function. ECHOES is complementary rather than a replacement: PIP is a pair-level weighting that still requires the analyst to express each statistic as a weighted pair (or n-tuple) count, whereas ECHOES returns explicit equal-weight point catalogs on which any estimator — pairwise or not — runs directly. Its distinct contributions are that it also restores redshift failures and imaging-systematic density modulations in the same completion, and that it stores the result cosmology-free; the fiber-collision piece alone is well served by PIP.

SIMBIG is a useful modern comparison point. It uses BOSS CMASS-South-like cuts and a forward-modeling program to connect nonlinear galaxy clustering statistics to cosmological parameters (C. Hahn et al. 2023a,b, 2024a,b). Later work extends the same style of analysis to summaries such as the wavelet scattering transform and marked power spectra (B. Régaldo-Saint Blancard et al. 2024; E. Massara et al. 2025). The scientific ambition is close to ours–to use information beyond the Gaussian two-point limit–but the observational interface remains the published BOSS selection and weighting system. Our product is complementary: it provides an ensemble of completed data catalogs so the observational correction can be sampled once and then reused across many statistics.

The closest precursor in this "ensemble of completed catalogs" sense is the DESI alternate-realizations (`altmtl`) program (J. Lasker et al. 2025), which reruns the fiber-assignment algorithm many times to produce an ensemble of realizations whose averaged pair counts give unbiased clustering — effectively a Monte-Carlo realization of the PIP probabilities. ECHOES shares the ensemble-of-realizations philosophy but differs in interface and scope: where `altmtl` resamples the *assignment* of a fixed target list (and is designed for two-point/PIP estimation), ECHOES resamples the *redshifts and positions* of the spectroscopically missing galaxies to yield equal-weight point catalogs intended for arbitrary statistics, and additionally folds in redshift-failure and imaging-systematic completion.

4.2. *Default KNN-field completion*

The default BOSS ECHOES product uses the KNN-field engine. The report's top-level numbers are as follows. The observed catalog has 109,636 CMASS-South galaxies. The real missing-target list contains 5,272 collision targets and 1,505 redshift-failure targets. The fixed base catalog therefore has 116,413 galaxies. With the imaging-systematic analog sampler enabled, seed 0 has 119,923 galaxies, including 3,510 analogs. Seeds 0–4 vary only at the $\simeq 0.1\%$ level in total count, as expected for the smooth analog step.

Figure 4 shows the restored-target population and sampled $n(z)$. Figure 5 compares the angular and two-dimensional clustering, and Figure 6 the projected correlation and redshift-space multipoles. The completed equal-weight catalogs reproduce the official weighted angular clustering to roughly 1.5% over the plotted angular range and the projected and multipole statistics at the 1–5% level, depending on scale and statistic. This is the expected accuracy target: the completed catalog is not an exact algebraic rewrite of the official weights, because it assigns redshifts and positions to missing galaxies rather than moving all their weight onto observed neighbors, but it should remain inside the systematic scale already tolerated by the reference BOSS treatment.

4.3. *Truth-known validation*

Two truth-known tests are used. The first hides the spectroscopic redshifts of a subset of real BOSS galaxies and then asks the completion machinery to recover the clustering of the original full catalog. The second applies the same machinery to independent MultiDark-Patchy mocks. In both tests, an "oracle" catalog that restores the hidden objects with their true redshifts provides a floor: if the oracle differs from truth, the difference comes from the masking and bookkeeping rather than from redshift assignment.

The main result is shown in Figure 7. The real-BOSS-truth mock and the Patchy mocks recover $w_p(r_p)$ to

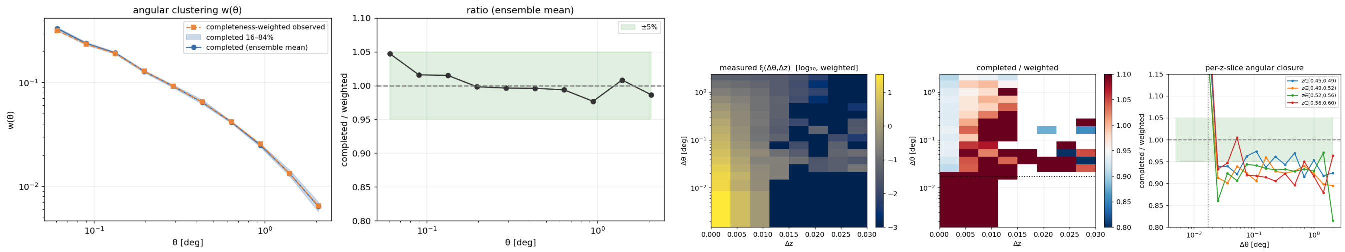

FIG. 5.— Two-point clustering of the equal-weight completed catalogs on the real BOSS CMASS-South data, with no completeness weights applied. *Left:* the angular correlation function $w(\theta)$ of the completed ensemble (mean and realization scatter, blue band) against the official w_c-weighted measurement (orange); the two agree to $\simeq 1.5\%$ across $0.05°$–$2°$, and the ratio panel shows no scale-dependent trend. *Right:* the two-dimensional correlation $\xi(\Delta\theta, \Delta z)$ measured directly in observed coordinates and its completed/weighted ratio across the plane. The anisotropy between angular and line-of-sight separation is reproduced to a few percent, so the geometric (Alcock–Paczynski) information carried by the full $\xi(\Delta\theta, \Delta z)$ plane is preserved without imposing a distance–redshift relation. The completion replaces survey weights by explicit equal-weight galaxies while leaving the measured two-point clustering unchanged at the percent level.

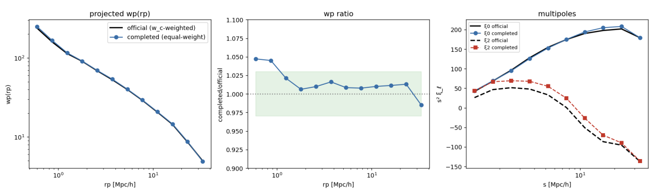

FIG. 6.— Standard redshift-space clustering of the equal-weight completed ensemble (no completeness weights) versus the official w_c-weighted BOSS measurement, with a Planck fiducial cosmology used only to convert redshift to distance. The projected correlation $w_p(r_p)$ (with the completed/official ratio) and the redshift-space monopole and quadrupole ξ_0, ξ_2 agree with the weighted reference at the 1–5% level over $0.5 < r_p < 40\,h^{-1}$Mpc. An equal-weight completed catalog is therefore a drop-in replacement for the weighted catalog in the standard two-point estimators.

about 1–2% over $0.5 < r_p < 40\,h^{-1}$Mpc. The oracle ratio lies in the range 0.997–1.005, so the remaining error is dominated by the redshift assignment posterior. The Patchy test is faithful above about $1\,h^{-1}$Mpc; the sub-Mpc deficit is a known one-halo limitation of the Patchy mocks rather than a failure mode inferred from real BOSS data.

4.4. *Higher-order and non-two-point checks*

The validation intentionally includes summaries that are nonlinear in the point set. For the nearest-neighbor distance CDFs (A. Banerjee & T. Abel 2021), the maximum absolute CDF differences between completed and truth catalogs are 0.0022, 0.0021, and 0.0011 for $k = 1, 2, 4$, respectively. For the counts-in-cells distribution (C. Uhlemann et al. 2020; O. Friedrich et al. 2020) in $8\,h^{-1}$Mpc cells, the mean is 0.660 versus 0.663 in truth, the variance-to-mean ratio is 2.37 versus 2.42, and the skewness is 2.90 versus 2.94. These tests are not exhaustive, but they are important because a product that only reproduces w_p could still fail for nearest-neighbor or cell-count statistics.

Because redshift failures, unlike fiber collisions, receive no close-pair anchor, we also check their assignment directly against the real-BOSS-truth redshifts, separated by missing kind. The recovered redshifts are unbiased for both populations — the median assigned-minus-true offset is -0.0008 for failures and $+0.0001$ for collisions, with comparable scatter ($\sigma \simeq 0.031$ versus 0.028) and catastrophic-outlier fractions ($|\Delta z| > 0.05$ of 10.7% versus 11.2%) — so the absence of a close-pair prior does not degrade the failures relative to collisions. The per-object ensemble is mildly over-confident for both kinds (the rank histogram of the true redshift departs from uniform at the ~ 0.1 Kolmogorov–Smirnov level), a small residual that is conservative for ensemble-mean statistics and bounded by the statistic-level calibration below.

4.5. *Calibration of completion uncertainty*

The completion posterior is intentionally narrower than a full survey posterior. It conditions on the observed CMASS-South realization and resamples only the unobserved spectroscopic information and smooth imaging analogs. In the Patchy calibration suite, the posterior uncertainty in $w_p(r_p)$ is 0.17–0.44% per bin, while sample variance over the same bins is 0.8–7.2%. The completion therefore adds only 6–36% of the sample-variance error in quadrature, depending on scale. The 68% completion band brackets the mock truth in only about 8% of bins because it is not intended to regenerate the large-scale density field. This is correct behavior: a user should add the completion covariance to a cosmic-variance or simulation covariance, not replace that covariance with the completion ensemble.

The same statement holds beyond the two-point function, which matters because the higher-order statistics are more sensitive to the small-scale redshift assignment. Across the Patchy suite we measure, per statistic bin, the realization-to-realization completion scatter and the mock-to-mock sample variance for the nearest-neighbor CDFs ($k = 1, 2, 4$) and the counts-in-cells variance-to-mean ratio. The median ratio of completion to sample-variance scatter is 0.28 for the kNN-CDF and 0.17 for counts-in-cells (Figure 10), so the completion remains a sub-dominant, addable uncertainty term for the non-Gaussian statistics as well, not only for w_p.

4.6. *KNN-field versus graphGP*

The KNN-field and graphGP routes agree on the main scientific conclusion but make different compromises. The KNN-field engine is sharper on the sub-Mpc collision scale because it directly uses nearby observed redshifts and the empirical close-pair prior. graphGP is smoother and more explicitly correlated across missing objects because all lines of sight share a density-field draw. In truth recovery, both routes recover the input clustering to the percent level. In the real-data head-to-head comparison, graphGP lies closer to the official weighted reference on larger projected scales, with a typical ratio near unity, while the KNN-field realization is modestly high in some bins. Because the KNN-field route is much cheaper and

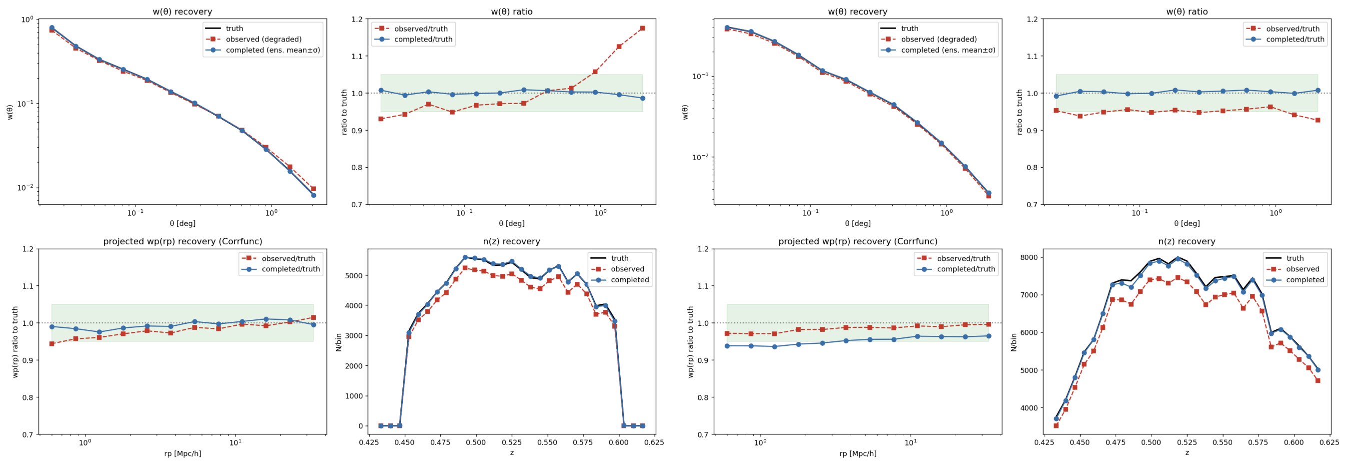

Fig. 7.— Truth-known recovery of the projected correlation $w_p(r_p)$, plotted as the ratio to the input truth. *Left:* a real-BOSS-truth test in which the spectroscopic redshifts of a subset of CMASS-South galaxies are hidden and then completed. The incomplete observed catalog is biased low at small r_p (missing close pairs); the completed ensemble recovers the truth to 1–2% over $0.5 < r_p < 40\,h^{-1}$Mpc. The oracle curve, which restores the hidden galaxies at their *true* redshifts, sets the achievable floor (ratio 0.997–1.005), so the residual is dominated by the redshift-assignment posterior rather than by the masking bookkeeping. *Right:* the same battery on independent MultiDark–Patchy mocks; recovery is faithful above $\simeq 1\,h^{-1}$Mpc, and the sub-Mpc deficit is a known one-halo limitation of those mocks, confirmed by the real-BOSS-truth panel which recovers the small scales. The completion reproduces an unobserved truth, not merely the weighted bookkeeping of the observed catalog.

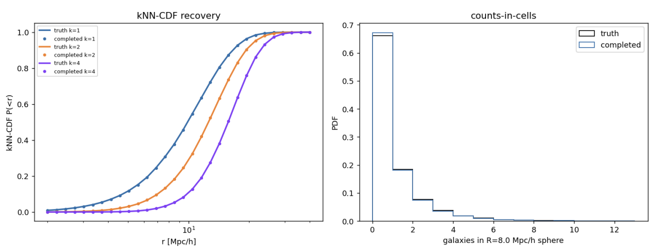

Fig. 8.— Higher-order, coincidence-sensitive validation of the completed catalogs against truth. *Left:* the nearest-neighbor (kNN) cumulative distribution functions for $k = 1, 2, 4$ — the distribution of distances from random points to the k-th nearest galaxy — which respond to duplicate points, missing close pairs, and redshift smearing. Completed and truth agree to a maximum $|\Delta\,\mathrm{CDF}| \leq 0.0022$, and the absence of a spike at small separation confirms that the local-analog imaging-systematic step introduces no exact duplicates. *Right:* the counts-in-cells distribution in $8\,h^{-1}$Mpc spheres; the full one-point density distribution is recovered (mean 0.660 versus 0.663, variance-to-mean 2.37 versus 2.42, skewness 2.90 versus 2.94), not merely its second moment.

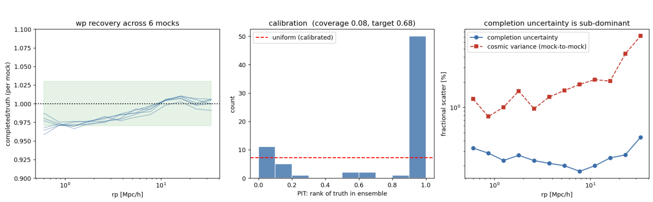

Fig. 9.— Calibration of the completion uncertainty against mock sample variance (MultiDark–Patchy). Per r_p bin, the realization-to-realization spread of the completed $w_p(r_p)$ (the completion uncertainty) is 0.17–0.44%, well below the 0.8–7.2% sample variance, so the completion adds only 6–36% of the sample-variance error in quadrature. Because the ensemble conditions on the single observed CMASS-South realization and resamples only the unobserved spectroscopic information, it is deliberately narrow and must be *added* to a cosmic-variance or simulation covariance, not used in place of it.

compresses to the compact public ECHOES posterior, it is the default release. graphGP is retained as a first-class alternative for surveys or statistics where correlated redshift-completion uncertainty is essential.

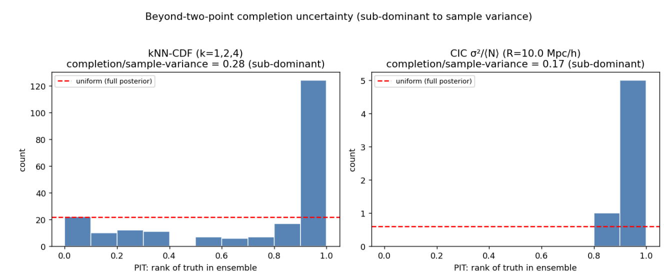

Fig. 10.— Beyond-two-point completion uncertainty. For the nearest-neighbor CDFs ($k = 1, 2, 4$, left) and the counts-in-cells variance-to-mean ratio (right), the completion (within-mock) scatter is a small fraction of the mock-to-mock sample variance — median ratios 0.28 and 0.17 — so the completion adds a sub-dominant non-Gaussian uncertainty. The PIT of the truth within the completion ensemble is intentionally non-uniform: as for w_p, the ensemble conditions on the single observed realization and does not regenerate the cosmic field, so its band is not meant to bracket the mock truth at the nominal rate.

Although the graphGP field is built in a fiducial comoving frame, that choice is a gauge rather than a prior. Repeating the redshift assignment under two extreme fiducial cosmologies — Planck ($\Omega_m = 0.315$) and Einstein–de Sitter ($\Omega_m = 1.0$) — changes the per-object redshifts by an RMS of only ~ 0.001 (correlation 0.9996), an order of magnitude below the intrinsic assignment scatter, and the recovered $w_p(r_p)$ by less than 0.1% (Figure 12). Because the kernel is measured from the data and the output is expressed in observed redshift, the graphGP route is data-driven and carries no cosmological prior.

5. SAMPLING THE COMPLETION POSTERIOR

For the default ECHOES release, a posterior sample is a completed catalog seed. The mean of a statistic over seeds estimates the statistic for the completed survey, and the covariance over seeds estimates uncertainty due to the completion. If $S[\mathcal{C}^{(s)}]$ is any statistic of a catalog,

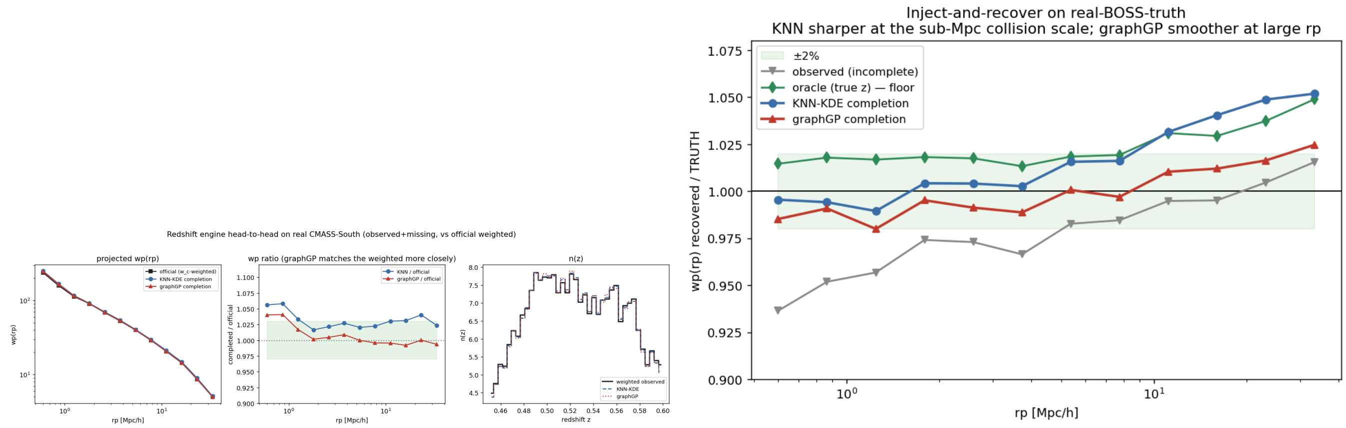

FIG. 11.— The default KNN-field redshift engine versus the graphGP conditional-field route, on the real CMASS-South data. *Left:* projected and angular clustering of the two completions against the official w_c-weighted reference; graphGP, being a smooth field draw, sits closer to the weighted reference on larger projected scales (ratio near unity), while the KNN-field realization is modestly high in some bins. *Right:* truth-known recovery of $w_p(r_p)$ for both engines; the KNN-field engine is sharper on the sub-Mpc fiber-collision scale because it uses nearby observed redshifts directly, whereas graphGP is smoother and slightly closer on larger scales. Both recover the input clustering to the percent level; neither dominates, and the cheaper KNN-field route — which compresses to the public posterior — is the default.

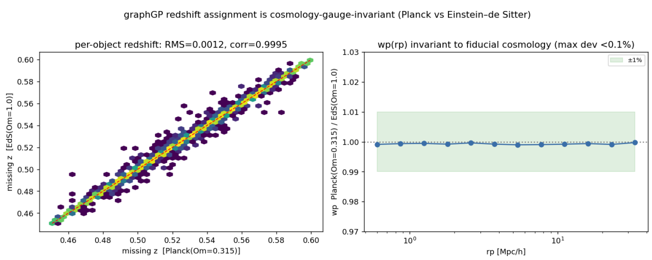

FIG. 12.— The graphGP redshift assignment is invariant to the internal fiducial cosmology. *Left:* per-object missing redshifts assigned under a Planck and an Einstein–de Sitter fiducial agree with RMS ~ 0.001 and correlation 0.9996, far below the intrinsic assignment scatter and the photo-z uncertainty. *Right:* the recovered $w_p(r_p)$ differs by less than 0.1% between the two fiducials. The fiducial distance–redshift relation is a coordinate (gauge) choice absorbed by the data-measured kernel, not a cosmological prior.

then

$$\bar{S} = \frac{1}{N_s}\sum_{s=1}^{N_s} S[\mathcal{C}^{(s)}], \tag{12}$$

$$C_{ab}^{\rm comp} = \frac{1}{N_s - 1}\sum_s \left(S_a[\mathcal{C}^{(s)}] - \bar{S}_a\right)\left(S_b[\mathcal{C}^{(s)}] - \bar{S}_b\right). \tag{13}$$

This covariance should be added to, or modeled jointly with, the ordinary survey covariance. It should not be interpreted as the covariance of the cosmological density field.

For graphGP the same equations apply, but the meaning of a seed changes. A seed selects a conditional density-field draw as well as the missing-galaxy redshift draws from Equation (11). The redshift assignments are therefore correlated across missing galaxies. This is the more faithful posterior object if the statistic is sensitive to coherent redshift uncertainty, but it is more expensive to construct and store. For the BOSS CMASS-South tests in this paper, the difference between graphGP and KNN-field is smaller than the sample-variance error budget for the standard clustering statistics.

The report evaluates several statistic-level responses. The angular $w(\theta)$ response to shifting the missing-galaxy redshift prior is far below the realization scatter, as expected for a mostly angular statistic. Radial statistics are more sensitive. The completed $w_p(r_p)$ correction uncertainty is at the 0.2–0.4% level, and the real-data comparison with official weighted BOSS clustering is at the 1–5% level across the displayed projected and multipole statistics. For nonlinear summaries, the nearest-neighbor CDF and counts-in-cells tests in Figure 8 show no large hidden failure from the local analogs or from the missing-redshift posterior. Users can repeat the same exercise for any proposed summary by running that summary on seeds $0, \ldots, N_s - 1$.

6. SYSTEMATICS, MASKS, AND TRUSTWORTHINESS

The completion also produces diagnostic maps rather than only final catalogs. The redshift-failure coupling to the local target density is consistent with zero in the current BOSS CMASS-South product, $h = -0.06 \pm 0.06$, while the fiber-collision coupling is detected, $h = -0.24 \pm 0.03$. This is physically reasonable: collisions occur in denser angular environments. The completed density map correlates with the total-target density at 0.98, and the trustworthiness map has a median scatter of 0.13. These diagnostics identify where the completion is extrapolating most strongly.

The optional mask-inpainting product fills 542 interior holes totaling 16.6 deg^2, about 0.9% of the footprint. The inpainting closure test is approximately 1% over 0.06°–2°. We keep this step optional because different analyses may prefer to leave bright-star holes and small mask features explicit in the random catalog. The default product therefore corrects spectroscopic incompleteness and smooth imaging weights while leaving the official footprint visible.

7. DATA AND CODE AVAILABILITY

The analysis code is publicly available at `https://github.com/yipihey/graphGP-cosmology`, and the interactive completion report at `https://yipihey.github.io/graphGP-cosmology/completion.html`. The compact ECHOES posterior, the matching random

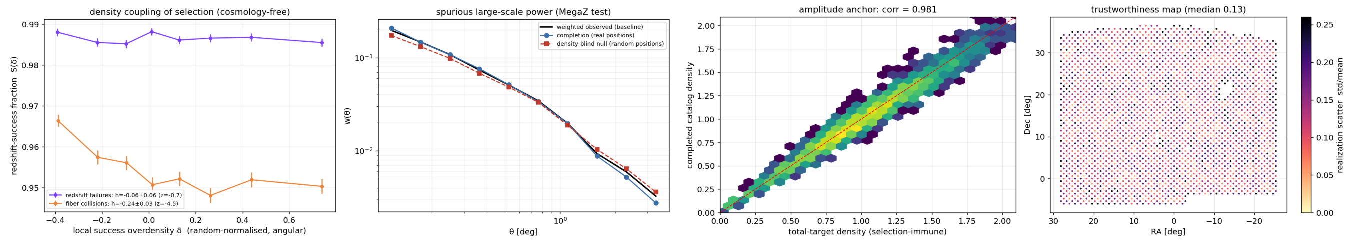

FIG. 13.— Completion diagnostics. *Left:* the spectroscopic success fraction as a function of the local target overdensity for the two missing-target kinds. Redshift failures are consistent with no density coupling ($h = -0.06 \pm 0.06$), whereas fiber collisions are coupled ($h = -0.24 \pm 0.03$) because close pairs occur in denser environments; since the completion places collided galaxies at their true imaging positions, this coupling is reproduced by construction rather than modeled. *Right:* the trustworthiness map — the galaxy-count scatter across completion realizations per sky cell (median 0.13) — which marks where the redshift posterior leaves the most freedom and the completion is extrapolating most strongly.

catalog, and a dependency-light sampler are released with the report; a reproducible ensemble of completed catalogs is specified by the set of integer seeds used. The file contents, the minimal sampling interface, and the relevant source modules are documented in Appendix C.

8. CONCLUSIONS

We have presented ECHOES, an equal-weight completed-catalog posterior for BOSS DR12 CMASS-South. The product restores spectroscopically missing galaxies at real SDSS imaging positions, samples their redshifts from a data-driven posterior, and represents smooth imaging-systematic weights as local analog galaxies. It therefore changes the user-facing data interface: instead of asking every new summary statistic to understand the BOSS completeness weights, an analyst can run the statistic directly on completed point catalogs and propagate observational-completion uncertainty by resampling seeds.

The validation is deliberately conservative. The completed ensemble recovers truth-known $w_p(r_p)$ to the 1–2% level over the validated scales, matches official weighted BOSS clustering at the few-percent level, reproduces nearest-neighbor CDFs and counts-in-cells diagnostics, and has a completion uncertainty smaller than the CMASS-South sample variance. The graphGP route provides an independent, correlated posterior construction and agrees with the default KNN-field route within the current BOSS error budget.

The broader implication is practical. Survey collaborations have learned how to publish two-point-ready data products. The next generation of analyses will not be limited to two-point functions, and the next generation of surveys will have more complicated assignment, calibration, and selection functions. A ECHOES posterior over completed survey catalogs is a way to preserve the historical division of labor–survey teams ground observational corrections in the data, theory teams test statistics and models–while making the interface useful for a broad range of summaries. DESI, Euclid, Roman, CSST, and Rubin/LSST will all face versions of this problem. BOSS CMASS-South is small enough to solve carefully and mature enough to validate against the standard answer; it is therefore a useful prototype for how future public survey products can serve both standard and beyond-two-point cosmology.

APPENDIX

REPRODUCIBLE ALGORITHM

This appendix writes the default KNN-field completion as an implementation recipe.

1. Load the BOSS DR12 CMASS-South LSS catalog with photometry and the matching random catalog. Apply the cuts RA $< 28°$ or RA $> 335°$, Dec $> -6°$, and $0.45 < z < 0.60$.

2. Convert SDSS `MODELFLUX` and `EXTINCTION` to magnitudes and colors. Train `PhotoZKNN(k=100)` on finite observed galaxies using features $(g-r, r-i, i-z, i)$.

3. Load the CMASS target catalog. Build the missing list by greedily tying no-spectrum targets to `WEIGHT_CP` hosts within $62''$ and `ZWARNING`$\neq 0$ targets to `WEIGHT_NOZ` hosts within $15'$.

4. Measure the observed close-pair Δz pool from pairs of observed galaxies within $62''$ and symmetrize it.

5. For each missing target, find the $K = 150$ angular nearest observed spectroscopic galaxies. On a 256-point redshift grid, form the product of the local field KDE with $\sigma_f = 0.004$, the photo-z KDE with $\sigma_p = 0.02$, and, for collisions, the close-pair prior.

6. Store each normalized posterior as a 65-level inverse CDF. Quantize the observed redshifts and inverse-CDF values to `uint16`; store positions as `float32`, `WEIGHT_SYSTOT` as `float16`, and provenance as `int8`.

7. To draw seed s, invert each missing target's CDF using uniform variates from `numpy.random.default_rng(s)`, add Gaussian redshift jitter $\sigma_f/2$, concatenate observed galaxies and restored targets, and add local imaging-systematic analogs with $1''$ angular jitter.

8. Measure statistics on many seeds. Use the seed mean as the completed survey statistic and the seed covariance as the completion covariance.

GRAPHGP IMPLEMENTATION DETAILS

The graphGP route uses the same observed BOSS catalog and missing target list but replaces the KNN line-of-sight density with a correlated posterior field. The implementation steps are:

1. Choose a fiducial distance-redshift relation for the internal field coordinate system and convert observed (α, δ, z) to $\mathbf{x}$.
2. Measure or tabulate the covariance $K_\xi(r)$ from the observed two-point function.
3. Build a sparse Vecchia neighbor graph over data and prediction points.
4. Form the FKP kernel field estimate (Equation 9) at the prediction points, and generate a ξ-correlated prior draw on the graph.
5. Combine them into the coverage-scaled ensemble draw (Equation 10).
6. Store $1 + \delta^{(s)}$ on a redshift-shell by HealPIX light cone.
7. For each missing target, evaluate Equation (11) on the same redshift grid used by the KNN-field sampler, then draw redshifts from the shared field realization.

The field posterior is not part of the default release because it is larger and more expensive, but it is implemented as a first-class alternative and was used for the comparisons in Section 4.6.

DATA PRODUCTS AND SAMPLER INTERFACE

The release in the report's `docs/data/` directory contains the compact posterior `cmass_south_posterior.npz` (the fixed observed catalog plus an inverse-CDF redshift posterior for each restored target, tabulated at 65 quantile levels and quantized to `uint16` for a 2.04 MB total), the matching uniform-footprint randoms `cmass_south_randoms.npz`, and a NumPy-only sampler `draw_samples.py`. A completed catalog is drawn from an integer seed with

```
from draw_samples import load_package, draw
pkg = load_package("cmass_south_posterior.npz")
cat = draw(pkg, seed=0)
```

which returns the arrays `ra`, `dec`, `z`, `prov`, and `N`; a reproducible ensemble is the set of seeds used. The relevant source modules are `twopt_density/boss.py` (BOSS loader), `cmass_targets.py` (missing-target recovery), `photoz.py` (the color-space kNN photo-z), `observed_ls.py` (the KNN-field completion and the graphGP redshift interface), `posterior_sampler.py` (the compact inverse-CDF posterior), `density_field.py` (graphGP posterior-field sampling), and `clustering_corrfunc.py` (the projected and multipole clustering measurements).

This manuscript draft is based on the BOSS completion products in the `graphGP-cosmology` repository. The BOSS data products are from SDSS-III.

SOFTWARE

Corrfunc (M. Sinha & L. H. Garrison 2020), NumPy, SciPy, Astropy, Healpy, graphGP

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